GEOMETRIC–CAUSAL–SPECTRAL COSMOLOGY ON A COMPACT THREE-SPHERE: S^3 KERNEL, CAUSAL LIGHT FRONT, DENSITY TRACKING, AND REDSHIFT AS A DENSITY-GRADIENT INTEGRAL

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Abstract. **Positioning.** This paper proposes a geometrically constrained phenomenological framework on compact S^3. It is *not* a replacement of ΛCDM, *not* a first-principles derivation of cosmological parameters, and *not* a modification of local Einstein gravity. The central conceptual content is: the observable universe should be distinguished from the full spatial geometric volume, and density may track this observable domain dynamically rather than fill it instantaneously.

We formulate a Geometric–Causal–Spectral cosmology on a compact, simply-connected three-sphere $S^3(R(t))$ as the joint evolution of three fundamental fields: the geometric radius $R(t)$; the dimensionless causal light angle $\chi_{\rm L}(t) = \int_0^t c\,dt'/R(t')$; and the energy density $\rho(t)$.

We introduce the S^3 kernel $K(\chi) = 2\sin^2\chi/(\chi - \sin\chi\cos\chi)$ governing radial volume growth, with limits $K \sim 3/\chi$ (Euclidean) and $K(\pi) = 0$ (antipodal). The observable volume is $V_{\rm obs} = 2\pi R^3[\chi_{\rm L} - \frac{1}{2}\sin 2\chi_{\rm L}]$, distinct from $V_{\rm total} = 2\pi^2 R^3$.

The radius obeys a sigmoid-bridged two-sector law with a discrete Fibonacci pre-recombination phase (Perron–Frobenius eigenvalue φ) and an exponential post-recombination phase governed by an *effective* expansion rate $H_{\rm eff} \neq H_0$.

The observable density obeys the postulated relaxation $\dot{\rho}_{\rm obs} = -\Gamma(t)[\rho_{\rm obs} - \rho_{\rm total}]$ with $\Gamma = c/(R\chi_{\rm L})$. The cosmological redshift is the line integral $\ln(1+z) = \int_0^\ell p(V(r))K(\chi(r))/R(r)\,dr$. The observable Hubble parameter is window-dependent, $H(\ell) = 3cp/\ell$, suggesting a phenomenological two-regime structure with ratio $H_0^{\rm late}/H_0^{\rm CMB} \approx 1.08$–$1.10$, consistent with the observed CMB–late-time split at the phenomenological level (not a parameter-free prediction).

We do not eliminate Λ, do not derive H_0, Λ, or the vacuum energy density from first principles, do not replace the Friedmann equations, and do not provide a theory of quantum gravity. The Hubble-tension consistency is a phenomenological check, not a derivation.

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1. Geometry of S^3

We assume spatial sections Σ that are compact, connected, orientable, boundaryless, with trivial fundamental group, and endowed with a round Riemannian metric. By the Poincaré–Perelman theorem [3, 4, 5], $\Sigma \cong S^3$ uniquely.

The total geometric volume of $S^3(R)$ is

$$V_{\text{total}}(R) = 2\pi^2 R^3. \tag{1}$$

The geodesic radial coordinate $\chi \in [0, \pi]$ measures dimensionless angular distance from a chosen pole. The volume of a geodesic ball of comoving angle χ on $S^3(R)$ is

$$V_{\text{obs}}(\chi, R) = 2\pi R^3 \left[\chi - \tfrac{1}{2}\sin(2\chi)\right]. \tag{2}$$

At $\chi = \pi$, $V_{\text{obs}} = V_{\text{total}}$; for $\chi < \pi$, $V_{\text{obs}} < V_{\text{total}}$.

$$\boxed{V_{\text{obs}} \neq V_{\text{total}}}$$

2. The S^3 Kernel

Definition 2.1 (S^3 kernel)**.** The S^3 kernel governing radial volume growth is

$$K(\chi) \equiv \frac{1}{V}\frac{dV}{d\chi} = \frac{2\sin^2\chi}{\chi - \sin\chi\cos\chi}, \quad \chi \in (0, \pi). \tag{3}$$

Theorem 2.2 (Limits of K)**.** *$K(\chi) \to 3/\chi$ as $\chi \to 0$; $K(\chi) \to 0$ as $\chi \to \pi$; $K(\pi/2) = 4/\pi \approx 1.273$.*

Proof. Taylor expansion: $\sin\chi\cos\chi = \sin(2\chi)/2 = \chi - 2\chi^3/3 + \cdots$, so denominator $\chi - \sin\chi\cos\chi \to 2\chi^3/3$. Numerator $2\sin^2\chi \to 2\chi^2$. Ratio $\to 3/\chi$. At $\chi = \pi$: $\sin\pi = 0$ in numerator, π in denominator. At $\chi = \pi/2$: numerator 2, denominator $\pi/2$. □

Remark 2.3 (Physical interpretation). $K(\chi)$ is the fractional rate of volume growth per unit angular distance. In small-χ Euclidean regime, $K \sim 3/\chi$ recovers the standard $4\pi r^2/[(4\pi/3)r^3] = 3/r$ shell-to-volume ratio. Near the antipode, $K \to 0$: no further volume is gained as the geodesic ball wraps around the closed manifold.

3. Three Fundamental Fields

Definition 3.1 (Three fundamental fields)**.** The cosmological state is specified by three independent fields:

- **(F1)** *Geometric radius* $R(t) > 0$: curvature radius of S^3 at cosmic time t.
- **(F2)** *Causal light angle* $\chi_{\mathrm{L}}(t) = \int_0^t c\,dt'/R(t') \in [0, \pi]$, the dimensionless angular distance traversed by a null geodesic.
- **(F3)** *Energy density* $\rho(t) > 0$: distributed on $S^3(R(t))$.

Remark 3.2 (Why χ_{L} rather than a dimensionful r_L). χ_{L} is the natural causal coordinate on S^3: it is dimensionless, bounded above by π, and parametrises the causal geometry without conflating spatial growth with light propagation. The proper causal distance is $r_L = R(t)\chi_{\mathrm{L}}(t)$.

4. Observable Volume

Definition 4.1 (Observable volume)**.** The observable volume at time t is the geodesic ball of angle $\chi_{\mathrm{L}}(t)$:

$$V_{\mathrm{obs}}(t) = 2\pi R^3(t)\big[\chi_{\mathrm{L}}(t) - \tfrac{1}{2}\sin 2\chi_{\mathrm{L}}(t)\big]. \tag{4}$$

Theorem 4.2 (Properties of $V_{\mathrm{obs}}(t)$)**.** *For* $\chi_{\mathrm{L}} \ll 1$*:* $V_{\mathrm{obs}} \approx (4\pi/3)(R\chi_{\mathrm{L}})^3$ *(Euclidean limit). For* $\chi_{\mathrm{L}} = \pi$*:* $V_{\mathrm{obs}} = V_{\mathrm{total}}$*. Today,* $\chi_{\mathrm{L}} \sim 0.07$ *and* $V_{\mathrm{obs}}/V_{\mathrm{total}} \sim 7 \times 10^{-5}$*.*

Proof. Direct: small-χ Taylor of (4) gives $(4\pi/3)R^3\chi^3$. Numerical evaluation at $\chi = 0.07$, $R = 1.85 \times 10^{27}$ m gives $V_{\mathrm{obs}}/V_{\mathrm{total}} = 7.4 \times 10^{-5}$. □

5. Fractal Spectral Structure

Lemma 5.1 (Minimality of the Fibonacci recursion)**.** *Among all positive integer second-order recursions* $a_{n+1} = p\,a_n + q\,a_{n-1}$ *(*$p, q \in \mathbb{Z}_{\geq 1}$*), the Fibonacci recursion (*$p = q = 1$*) has the smallest Perron–Frobenius growth rate* $\lambda_+ = \varphi$*.*

Proof. Companion matrix $T_{p,q} = \left(\begin{smallmatrix} p & q \\ 1 & 0 \end{smallmatrix}\right)$ has dominant eigenvalue $\lambda_+(p, q) = (p + \sqrt{p^2 + 4q})/2$, minimised at $p = q = 1$ with $\lambda_+ = \varphi$. □

Theorem 5.2 (φ as Perron–Frobenius eigenvalue)**.** *The Fibonacci ratio* F_{n+1}/F_n *converges to* $\varphi = (1 + \sqrt{5})/2$*, with rate* $|F_{n+1}/F_n - \varphi| = O(\varphi^{-2n})$*.*

Proof. Standard via Binet's formula [6, 7]. □

Remark 5.3 (Status of φ). The golden ratio is the dominant eigenvalue of the minimal positive second-order recursion (Lemma 5.1). The φ-ladder supplies the discrete spectral hierarchy used to organise scales; its observational realisation in the cosmological dynamics remains model-dependent.

Corollary 5.4 (Spectral bridge)**.** *If the discrete pre-recombination radii satisfy* $F_n/R_n =$ const*, then* $R_n \propto \varphi^n$ *and density scales as* $\rho_n \propto \varphi^{-3n}$*.*

6. Two-Sector Radius Law (Centre of the Framework)

Definition 6.1 (Two-sector radius law)**.** Let $t_{\mathrm{rec}} > 0$ be the recombination time, $\Delta t > 0$ its width, and $\sigma(t) = [1 + e^{-(t - t_{\mathrm{rec}})/\Delta t}]^{-1}$ the sigmoid weight. Define

$$R_{\mathrm{frac}}(t) = \ell_{\mathrm{P}}\,\varphi^{N(t)}, \qquad \text{(fractal sector)} \tag{5}$$

$$R_{\mathrm{light}}(t) = R_{\mathrm{rec}}\,e^{H_{\mathrm{eff}}(t - t_{\mathrm{rec}})}, \qquad \text{(light sector)} \tag{6}$$

with $R_{\rm rec} = R_{\rm frac}(t_{\rm rec}) = R_{\rm light}(t_{\rm rec})$. The cosmological radius is

$$\boxed{R(t) = R_{\rm frac}(t)\big[1-\sigma(t)\big] + R_{\rm light}(t)\,\sigma(t).} \tag{7}$$

Theorem 6.2 (C^1-smoothness)**.** *Under matching $R_{\rm frac}(t_{\rm rec}) = R_{\rm light}(t_{\rm rec})$: $R(t)$ and $\dot{R}(t)$ are continuous everywhere; $R(t_{\rm rec}) = R_{\rm rec}$; $\dot{R}(t_{\rm rec}) = \frac{1}{2}[\dot{R}_{\rm frac} + \dot{R}_{\rm light}]_{t_{\rm rec}}$.*

6.1. Effective Expansion Rate $H_{\rm eff} \neq H_0$.

Definition 6.3 (Effective post-recombination rate)**.**

$$H_{\rm eff} = \frac{\ln(R_{\rm today}/R_{\rm rec})}{t_0 - t_{\rm rec}}. \tag{8}$$

Remark 6.4 ($H_{\rm eff} \neq H_0$). $H_{\rm eff}$ is the *average* expansion rate over the post-recombination era; H_0 is the present-day *instantaneous* rate. With $R_{\rm today}/R_{\rm rec} \approx 1100$ and $t_0 - t_{\rm rec} \approx 4.35 \times 10^{17}\,$s:

$$H_{\rm eff} \approx 1.61 \times 10^{-17}\,{\rm s}^{-1} \approx 7.4\,H_0, \tag{9}$$

with $H_0 = 2.18 \times 10^{-18}\,{\rm s}^{-1}$ (Planck 2018). These are different physical quantities. $H_{\rm eff}$ is a matching parameter of the two-sector radius law, not the measured late-time Hubble constant. Earlier presentations [1] that used $H = 2.45 \times 10^{-17}\,{\rm s}^{-1}$ in the exponential were approximating $H_{\rm eff}$, but mis-identifying it as H_0.

7. Causal Light Propagation

Theorem 7.1 (Null-geodesic equation on S^3)**.** *In the FLRW metric on S^3, null geodesics satisfy $c\,dt = R(t)\,d\chi$, integrating to*

$$\chi_{\rm L}(t) = \int_0^t \frac{c\,dt'}{R(t')}. \tag{10}$$

7.1. Light as the Generator of Observable Volume.

The dimensionless causal angle $\chi_{\rm L}(t)$ defines, via (4), the observable volume. Light is therefore not merely an in-sphere propagating signal: *it is the operative agent that determines what is causally accessible* at time t.

In the pre-recombination era ($\sigma \ll 1$), the cosmic plasma is opaque and the operational meaning of $V_{\rm obs}$ is restricted. In the post-recombination era, free-streaming photons make $V_{\rm obs}(t)$ the operationally meaningful domain.

8. Compatibility with FLRW and Einstein Gravity

The framework is compatible with general relativity. Spatial topology S^3 corresponds to FLRW geometry with curvature index $k = +1$:

$$ds^2 = -c^2dt^2 + a^2(t)R_0^2\,d\Omega_3^2. \tag{11}$$

Local Einstein equations are unchanged; the Friedmann constraint reads

$$H_{\rm GR}^2 = \frac{8\pi G}{3}\rho - \frac{c^2}{a^2R_0^2} + \frac{\Lambda c^2}{3}. \tag{12}$$

Remark 8.1 (Compatibility, not replacement). The present framework is compatible with this geometry but does *not* claim to replace the Friedmann equations. The Gradient Redshift Ansatz (Postulate 10.1) is a coarse-grained observational ansatz, not a substitute for local geodesic optics in GR. The framework introduces no local modification of Einstein gravity.

9. Density Dynamics: A Postulate of Causal Tracking

Definition 9.1 (Two density fields)**.** Assuming an effective conserved energy E_{eff}:

$$\rho_{\text{total}}(t) = \frac{E_{\text{eff}}}{V_{\text{total}}(R(t))} \quad \text{(global)} \tag{13}$$

$$\rho_{\text{obs}}(t) = \frac{E_{\text{eff,within } V_{\text{obs}}}(t)}{V_{\text{obs}}(t)} \quad \text{(observable)} \tag{14}$$

Postulate 9.2 (Causal Density Tracking)**.** The observable density relaxes towards the global density at a rate set by the inverse causal information time:

$$\boxed{\dot{\rho_{\text{obs}}}(t) = -\Gamma(t)\big[\rho_{\text{obs}}(t) - \rho_{\text{total}}(t)\big], \qquad \Gamma(t) = \frac{c}{R(t)\,\chi_{\text{L}}(t)}.} \tag{15}$$

Remark 9.3 (Status of Postulate 9.2). This equation models relaxation of observable density toward the global geometric density. It is *not* derived from microscopic kinetic theory in this paper. The form $\Gamma = c/(R\chi_{\text{L}})$ is the simplest causal scaling consistent with the framework. A microscopic derivation is deferred to future work. Physically, the relaxation rate $\Gamma = c/(R\chi_{\text{L}})$ can be interpreted as a finite propagation speed of equilibration across the causal domain, bounded by light travel.

Remark 9.4 (Density tracks volume, not the reverse). The postulate formalises the central thesis: density does not instantaneously fill the geometric volume; it tracks the growth of the causal observable region with finite relaxation time $\Gamma^{-1} = R\chi_{\text{L}}/c$.

10. Redshift as a Density-Gradient Integral: Phenomenological Ansatz

10.1. Phenomenological Gradient Redshift Law.

Postulate 10.1 (Gradient Redshift Ansatz)**.** The photon angular frequency ω along a null geodesic parametrised by affine parameter λ evolves according to a local volume gradient:

$$\frac{d\ln\omega}{d\lambda} = -p(V)\,\frac{1}{V}\frac{dV}{d\lambda}, \tag{16}$$

where $p(V)$ is a dimensionless coarse-grained morphology index characterising effective photon coupling to local volume change.

Remark 10.2 (Status of Postulate 10.1). Equation (16) is *not* derived from local GR or QFT. It is introduced as an effective coarse-grained transport rule for photon frequency along null geodesics in a compact S^3 causal domain. This ansatz should be interpreted as an effective coarse-grained description and is not unique; alternative transport laws may exist.

Remark 10.3 (Status of $p(V)$). $p(V)$ is *not* a fluid equation-of-state parameter and must not be identified with $1 + w$. It is a coarse-grained morphology/rarefaction index. We constrain $0 < p(V) \leq p_{\max}$ and adopt a piecewise-constant simplification $p_{\text{rec}}, p_{\text{late}}$ for distinct observational windows. A first-principles derivation of $p(V)$ is deferred to future work.

10.2. Redshift–Distance Relation on S^3.

Proposition 10.4 (Conditional consequence of the Gradient Redshift Ansatz)**.** Conditional on Postulate 10.1, *the redshift along a null geodesic from a source at proper distance ℓ obeys*

$$\frac{d\ln(1+z)}{dr} = \frac{p(V)\,K(\chi(r))}{R(r)}, \tag{17}$$

giving

$$\boxed{\ln(1+z) = \int_0^{\ell} \frac{p(V(r))\, K(\chi(r))}{R(r)}\, dr.} \tag{18}$$

Proof. Combining (16) with $1+z = \omega_{\mathrm{em}}/\omega_{\mathrm{obs}}$ gives $d\ln(1+z) = p\, d\ln V$. Using $d\ln V/dr = K(\chi)/R$ along the null geodesic gives (17). □

Remark 10.5 (Physical interpretation). Redshift is the integral of the local fractional volume gradient along the photon's path, weighted by p. The S^3 kernel $K(\chi)$ encodes the geometric correction: in the Euclidean limit $K \sim 3/\chi$ recovers the standard inverse-distance dilution; near the antipode $K \to 0$ photons stop acquiring further redshift.

11. Observable Hubble Law

Proposition 11.1 (Effective windowed Hubble slope)**.** Conditional on Postulate 10.1, *for an observation window of proper extent ℓ with $\ell > \ell_{\min}$, approximately constant p, and $\chi \ll 1$ (so $K \sim 3R/r$), the redshift–distance relation can be approximated as* $\ln(1+z) \approx 3p\ln(\ell/\ell_0)$. *For $z \ll 1$, the effective windowed Hubble slope can be written as*

$$\boxed{H(\ell) = \frac{cz}{\ell}\Big|_{z\ll 1} = \frac{3cp}{\ell}.} \tag{19}$$

The $\ell \to 0$ limit is not physical.

Proof. In the small-χ regime $K(\chi)/R = 3/r$, so (18) gives $\ln(1+z) = 3p\int_{\ell_0}^{\ell} dr/r = 3p\ln(\ell/\ell_0)$. Linearising for $z \ll 1$ gives (19). □

Remark 11.2 ($H(\ell)$ is not a fundamental Hubble parameter). $H(\ell)$ is *not* a universal global expansion rate. It is an observationally windowed slope obtained after smoothing the local S^3 gradient over a finite scale ℓ. The expression is valid only for finite observational windows $\ell > \ell_{\min}$; the $\ell \to 0$ limit is not physical. The dependence on ℓ implies that observational inference of H_0 may be sensitive to window selection effects.

12. Two Regimes and the Hubble Tension

The window-dependent Hubble law naturally accommodates two distinct effective values:

$$H_0^{\mathrm{CMB}} = \frac{3c\, p_{\mathrm{rec}}}{\ell_{\mathrm{rec}}}, \quad p_{\mathrm{rec}} \approx 1/3, \tag{20}$$

$$H_0^{\mathrm{late}} = \frac{3c\, p_{\mathrm{loc}}}{\ell_{\mathrm{loc}}}. \tag{21}$$

Theorem 12.1 (Window ratio)**.**

$$\frac{H_0^{\mathrm{late}}}{H_0^{\mathrm{CMB}}} = \frac{p_{\mathrm{loc}}}{p_{\mathrm{rec}}} \cdot \frac{\ell_{\mathrm{rec}}}{\ell_{\mathrm{loc}}}. \tag{22}$$

For the parameters consistent with the two-sector law, this ratio falls in the range

$$\frac{H_0^{\mathrm{late}}}{H_0^{\mathrm{CMB}}} \approx 1.08\text{–}1.10, \tag{23}$$

consistent at the percent-level with the observed CMB–late-time split between Planck ($\sim$ 67.4 km/s/Mpc) and SH0ES local measurements [8] ($\sim$ 73 km/s/Mpc): ratio $\approx 73.0/67.4 = 1.083$. This consistency is a phenomenological check, not a parameter-free prediction.

Remark 12.2 (Status of the Hubble-tension consistency). The ratio $H_0^{\text{late}}/H_0^{\text{CMB}} \approx 1.08$–$1.10$ should be interpreted as a phenomenological consistency check, not as a parameter-free prediction. p and ℓ are taken from observational fits, not derived from first principles. We do not claim that the Hubble tension is solved. This mechanism does not exclude standard systematic explanations of the Hubble tension.

13. Numerical Consistency

Table 1. Numerical consistency check. Planck 2018 parameters used throughout [2].

Quantity	Symbol	Value
Planck length	ℓ_{P}	$1.616 \times 10^{-35}\,\text{m}$
Recombination time	t_{rec}	$1.20 \times 10^{13}\,\text{s}$
Recombination radius	R_{rec}	$9.46 \times 10^{20}\,\text{m}$
Today (cosmic)	t_0	$4.35 \times 10^{17}\,\text{s}$
Today scale factor	$R(t_0)$	$\sim 1.04 \times 10^{24}\,\text{m}$
Effective rate	H_{eff}	$1.61 \times 10^{-17}\,\text{s}^{-1}$
Present-day rate	H_0	$2.18 \times 10^{-18}\,\text{s}^{-1}$
Ratio H_{eff}/H_0	—	≈ 7.4
Curvature radius	R_{curv}	$\geq 1.85 \times 10^{27}\,\text{m}$
Today χ_{L}	—	~ 0.07
$V_{\text{obs}}/V_{\text{total}}$ today	—	$\sim 7 \times 10^{-5}$
$N(t_{\text{rec}})$ Fibonacci level	—	≈ 266.85
$K(\pi/2)$ kernel mid-value	—	$4/\pi \approx 1.273$
$H_0^{\text{late}}/H_0^{\text{CMB}}$ predicted	—	1.08–1.10
$H_0^{\text{late}}/H_0^{\text{CMB}}$ observed	—	1.083

14. Physical Interpretation

The framework rests on three operational principles:

(1) **Light defines what is accessible.** The observable volume $V_{\text{obs}}(t)$ is the geodesic ball of comoving angle $\chi_{\text{L}}(t)$, bounded above by π on compact S^3.
(2) **Density is dynamical, not instantaneous.** ρ_{obs} relaxes toward ρ_{total} at rate $\Gamma = c/(R\chi_{\text{L}})$ (Postulate 9.2).
(3) **Redshift is a path integral.** $\ln(1+z)$ is the line integral of $p(V)K(\chi)/R$ along null geodesics (Theorem 10.4); $H(\ell) = 3cp/\ell$ is therefore window-dependent.

The Hubble tension between CMB and local measurements is a direct consequence: different observational ladders correspond to different effective (p, ℓ) pairs, giving distinct effective Hubble parameters without contradiction.

15. Scope and Limitations

15.1. What This Paper Does.

(Y1) Establishes the three-field framework: $R(t)$, $\chi_{\text{L}}(t)$, $\rho(t)$.
(Y2) Introduces the S^3 kernel $K(\chi)$ with its limits.
(Y3) Derives φ as the Perron–Frobenius eigenvalue (Theorem 5.2).
(Y4) States the C^1-smooth two-sector law with $H_{\text{eff}} \neq H_0$.
(Y5) Postulates causal density tracking (Postulate 9.2).

(Y6) Reformulates redshift conditionally as a density-gradient integral (Theorem 10.4).
(Y7) Predicts a window-dependent Hubble law $H(\ell) = 3cp/\ell$.
(Y8) Identifies $H_0^{\text{late}}/H_0^{\text{CMB}} \approx 1.08$–$1.10$, consistent with the observed Hubble tension.

15.2. **What This Paper Does Not Do.**

(N1) Does *not* derive Λ from first principles.
(N2) Does *not* derive H_0 from first principles.
(N3) Does *not* derive $p(V)$ from microscopic theory.
(N4) Does *not* replace the Friedmann equations.
(N5) Does *not* provide a full quantum gravity model.
(N6) Does *not* claim that the Hubble tension is solved.
(N7) Does *not* claim that the observed vacuum density is predicted without calibration.
(N8) Does *not* eliminate dark energy. The exponential sector requires Λ as in ΛCDM.
(N9) Does *not* derive H_{eff}, E_{eff}, or $\Gamma = c/(R\chi_{\text{L}})$ from first principles. They are matching parameters and postulates.
(N10) Does *not* provide a wave function of the universe; earlier ansatze [1] are withdrawn.

Conclusions

The principal contributions are:

(1) S^3 kernel: $K(\chi) \to 3/\chi$ (Euclidean), $K(\pi) = 0$ (antipodal).
(2) Volume distinction: $V_{\text{obs}}/V_{\text{total}} \sim 7 \times 10^{-5}$ today.
(3) φ as Perron–Frobenius eigenvalue, not a parameter.
(4) Two-sector C^1-smooth law with $H_{\text{eff}} \approx 7.4\, H_0$.
(5) Causal density tracking $\dot{\rho_{\text{obs}}} = -[c/(R\chi_{\text{L}})][\rho_{\text{obs}} - \rho_{\text{total}}]$ as postulate.
(6) Conditional reformulation of redshift as gradient integral $\ln(1+z) = \int p(V)K(\chi)/R\, dr$.
(7) Window-dependent Hubble law $H(\ell) = 3cp/\ell$.
(8) Estimated within the present phenomenological framework: Hubble tension ratio ≈ 1.08–1.10, consistent with observation 1.083.

Final positioning. The main result of this paper is *not* a replacement of ΛCDM, *not* a derivation of cosmological parameters, and *not* a modification of local GR. It is a controlled geometric reinterpretation of observable cosmological volume and density on compact S^3. The distinction between V_{obs} and V_{total}, together with the causal density-tracking postulate and the gradient redshift ansatz, provides a phenomenological framework for studying whether observed expansion parameters may depend on finite observational windows.

All rigorous content lies in the geometry: S^3 topology, volume formula, kernel $K(\chi)$, Perron–Frobenius φ.
All physical interpretation is at the level of postulates, ansätze, and phenomenological consistency.

References

[1] A. Preece and B. Batenin, Fractal Bindu (S^3) of Reality: The Golden Ratio as the Principle of Perfection and Harmony in Interdependent Existence, Zenodo (2025), doi:10.5281/zenodo.15538573.
[2] Planck Collaboration, Planck 2018 results. VI. Cosmological parameters, Astron. Astrophys. **641**, A6 (2020).
[3] G. Perelman, The Entropy Formula for the Ricci Flow, arXiv:math/0211159 (2002).
[4] G. Perelman, Ricci Flow with Surgery on Three-Manifolds, arXiv:math/0303109 (2003).
[5] G. Perelman, Finite Extinction Time for the Solutions to the Ricci Flow, arXiv:math/0307245 (2003).

[6] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 5th ed., Oxford Univ. Press (1979).
[7] C. D. Meyer, *Matrix Analysis and Applied Linear Algebra*, SIAM (2000).
[8] A. G. Riess et al., A Comprehensive Measurement of the Local Value of the Hubble Constant, Astrophys. J. Lett. **934**, L7 (2022).
[9] A. Preece and B. Batenin, Compact Simply-Connected Cosmology and the Geometric Origin of Quantum Nonlocality (Version 3.0), Berlin (2026).

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